Data Analytics Learning

Intro to Data Analytics

*Data Analytics is the process of examining datasets to draw conclusions, using tools and techniques from statistics, programming, and business intelligence. It involves:

*Data Collection & Cleaning

*Data Visualization

*Statistical Analysis

*Tools: Excel, SQL, Python (Pandas, NumPy, Matplotlib), R, Tableau, Power BI

*Advanced: Machine Learning, Big Data (Hadoop, Spark), Cloud Analytics.

Books (Core Knowledge)

*Practical Statistics for Data Scientists

Author/Publisher: O'Reilly (Peter Bruce & Andrew Bruce)

Link: http://103.203.175.90:81/fdScript/RootOfEBooks/E%20Book%20collection%20-%202023%20-%20E/CSE%20ITAIDSML/Practical_Statistics_for_Data_Scientist_by_Peter_Bruce,_Andrew_Bruce.pdf

*Python for Data Analysis

Author/Publisher: O'Reilly (Wes McKinney)

Link: https://fasttacks.com/wp-content/uploads/2024/07/Wes-McKinney-Python-for-Data-Analysis_-Data-Wrangling-with-pandas-NumPy-and-Jupyter-OReilly-Media-2022.pdf

*Storytelling with Data

Author: Cole Nussbaumer Knaflic

Link: https://www.wiley.com/en-us/Storytelling+with+Data%3A+A+Data+Visualization+Guide+for+Business+Professionals-p-9781119002253

*The Data Warehouse Toolkit

Author: Ralph Kimball

Link: https://www.kimballgroup.com/data-warehouse-business-intelligence-resources/books/data-warehouse-dw-toolkit/

Notes (Documentation & Learning Paths)

W3Schools SQL Tutorial

Tool/Focus: SQL

Link: https://www.youtube.com/playlist?list=PLxSn8YMF02XWIQdf-c07-MxSjgpO6eySi

Pandas Documentation

Tool/Focus: Python Library (Data Manipulation)

Link: https://pandas.pydata.org/docs/

NumPy Documentation

Tool/Focus: Python Library (Numerical Computing)

Link: https://docs.scipy.org/

Tableau Public Resources

Tool/Focus: Data Visualization

Link: https://community.tableau.com/s/news/a0A8b00002JfZQhEAN/tableau-desktop-resources-for-newbies

Power BI Learning Path

Tool/Focus: Microsoft BI Tool

Link: https://learn.microsoft.com/en-us/training/paths/model-data-power-bi/

One-shots (Quick Courses/Tutorials)

*Data Analytics in 1 Hour

Provider: Simplilearn

Link: https://www.youtube.com/watch?v=rw6m1mMWSQA

*Excel Data Analysis Crash Course

Provider: Leila Gharani

Link: https://www.youtube.com/channel/UCJtUOos_MwJa_Ewii-R3cJA

*Power BI Crash Course

Provider: Pragmatic Works

Link: https://www.youtube.com/pragmaticworks

*Tableau Crash Course

Provider: Edureka

Link: https://www.youtube.com/watch?v=cSuo9uy7_Kk

Courses / Playlists

Free Courses

*FreeCodeCamp – Data Analysis with Python (Full Course)

Link: https://www.freecodecamp.org/learn/data-analysis-with-python

*Simplilearn – Data Analytics Full Course (8 hrs)

Link: https://www.youtube.com/watch?v=a3urSV7q44I

*Kaggle Micro-Courses

Focus: Python, Pandas, Data Viz, etc.

Link: https://www.kaggle.com/general/198508

*Microsoft Learn – Data Analytics

Focus: Power BI

Link: https://learn.microsoft.com/en-us/training/paths/data-analytics-microsoft/

Paid Courses

*Coursera – Google Data Analytics Professional Certificate

Link: https://www.coursera.org/professional-certificates/google-data-analytics

*Coursera – IBM Data Analyst Professional Certificate

Focus: Excel and R

Link: https://www.coursera.org/professional-certificates/ibm-data-analyst-r-excel

*Udemy – Data Analytics Bootcamp

Link: https://www.udemy.com/topic/data-analysis/

(Topic page with various course options)

Roadmap

■ Roadmap for Data Analysis (Detailed One-Page)

1. Foundations

- Math & Stats: Descriptive stats, probability, hypothesis testing, regression basics
- Excel/Sheets: Cleaning, formulas, pivot tables, dashboards

2. Programming Basics

- Python (pandas, numpy, matplotlib, seaborn) or R (dplyr, ggplot2)
- Focus on manipulation, visualization, automation

3. Databases & SQL

- SQL: SELECT, WHERE, JOIN, GROUP BY
- Optimization, MySQL, PostgreSQL

4. Data Cleaning & Wrangling

- Handle missing values, outliers, duplicates
- APIs, JSON/CSV/Excel, pandas/dplyr transformations

5. Data Visualization

- Python: matplotlib, seaborn, plotly | R: ggplot2
- Tools: Tableau, Power BI, Google Data Studio

6. Advanced Statistics & Analytics

- Regression, ANOVA, Chi-square
- Time series basics, A/B testing

7. Business / Domain Knowledge

- Finance, Marketing, Healthcare
- KPIs, business metrics

8. Advanced Tools & Big Data

- BigQuery, AWS Redshift, Azure Synapse
- Spark, Hadoop, ETL, Data Warehousing

9. Machine Learning (Optional)

- Regression, Classification, Clustering, PCA
- Scikit-learn basics

10. Projects & Portfolio

- Datasets: Kaggle, UCI ML repo
- Projects: Dashboards, Churn analysis, Forecasting
- Showcase on GitHub & LinkedIn